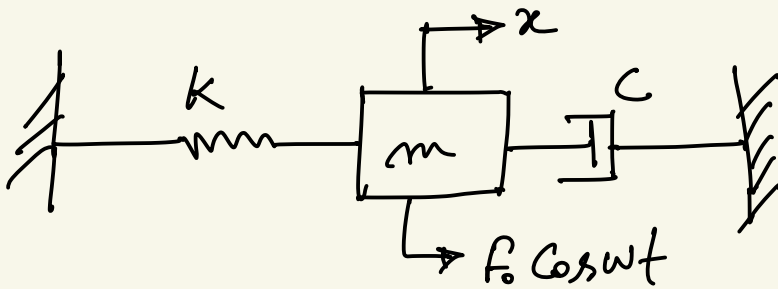




Frequency Response Analysis:

In our first lecture, we studied a spring mass damper system with a sinusoidal input.



$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$$

$$\Rightarrow \ddot{x} + 6.25 \dot{x} + 40x = 4.5 \cos 6t$$

The solution

$$x = x_c + x_p$$

x_c = Complementary solution
or natural solution
or transient solution

x_p = particular solution
= Steady state solution

$$x = \underbrace{e^{-3.125t} (-0.0127 \cos 5.4986t - 0.1366 \sin 5.4986t)}_{x_c} + \underbrace{(0.0127 \cos 6t + 0.1186 \sin 6t)}_{x_p}$$

As time goes by, ($t \rightarrow \infty$), $x_c \rightarrow 0$.
Only part that remains is x_p .

$$\begin{aligned}
 x = x_p &= 0.0127 \cos 6t + 0.1186 \sin 6t \\
 &= A \cos(6t + \varphi) \\
 &= \text{steady-state response.}
 \end{aligned}$$

As a result, in an LTI system, A sinusoidal input also generates a sinusoidal output. However, their amplitude is changed and there is a phase shift.

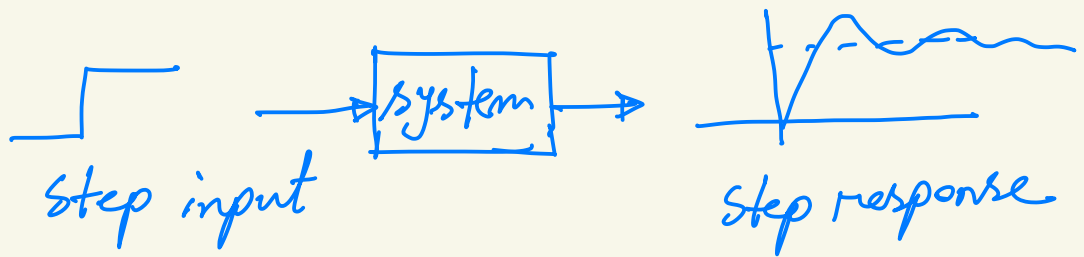
$$\begin{array}{c}
 \text{Input} \\
 \hline
 A_i \cos 6t
 \end{array}$$

$$\begin{array}{c}
 \text{Output} \\
 \hline
 A \cos(6t + \varphi)
 \end{array}$$

$$\begin{array}{c}
 A_i \neq A \\
 \text{and } \varphi > 0
 \end{array}$$

A frequency response approach characterizes the amplitude change and phase shift for different excitation frequencies.

So far we have studied the character of certain standard systems to certain simple inputs.



We can characterize t_p , t_s , t_r , M_p etc. $\rightarrow$ transient properties

We used PID controller to modify those.

We also studied steady-state error. We studied root locus for more complex systems.

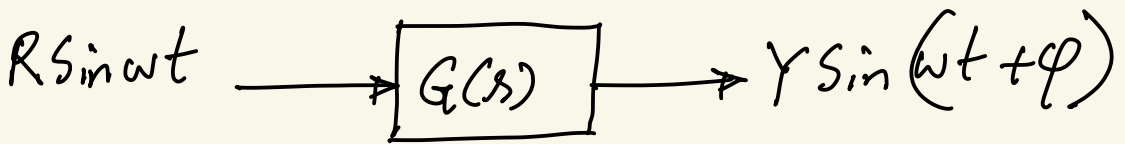
We saw that we can modify the behavior of root locus by using compensator.

However, with this time response approach, it is difficult to quantify the effect of higher order poles and zeros. And how the system

responds to complex inputs.

A frequency response approach can help us address these issues.

The idea of frequency response is that how the system response change as a function of input frequencies.



(Steady-state part)

we are interested in two quantities

1. Scaling or Amplitude ratio

$$\frac{Y}{R} = \frac{\text{output amplitude}}{\text{input amplitude}}$$

2. Phase shift

ϕ

We can obtain these two quantities by a "brute force" approach.

That is, we can try different input frequencies and record the output response. And obtain amplitude change versus frequency as well as phase change versus frequency.

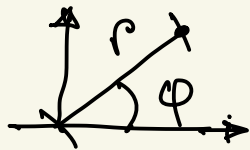
However, there is an easy and short-cut approach.

If we somehow obtain the transfer function of a system, and put $s = j\omega$, then we can obtain those two quantities of interest.

$G(s)$ is a complex number. $G(j\omega)$ is also a complex number.

$r = |G(j\omega)| \rightarrow \text{scaling / amplitude ratio}$

$$\angle G(j\omega) = \phi$$



This approach is important for designing controllers, filters, sensors, designing mechanical systems etc.

Let's consider the transfer function of a second order system.

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G(s) = \frac{1}{\left(\frac{s}{\omega_n}\right)^2 + 2\zeta \frac{s}{\omega_n} + 1}$$

put $s = j\omega$

$$G(j\omega) = \frac{1}{\left(\frac{j\omega}{\omega_n}\right)^r + 2\zeta \frac{j\omega}{\omega_n} + 1}$$

$$= \frac{1}{-\frac{\omega^r}{\omega_n^r} + 1 + j 2\zeta \frac{\omega}{\omega_n}}$$

$$G(j\omega) = \frac{1}{\left[1 - \left(\frac{\omega}{\omega_n}\right)^r\right] + j 2\zeta \frac{\omega}{\omega_n}}$$

$$|G(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^r\right]^2 + 4\zeta^2 \left(\frac{\omega}{\omega_n}\right)^r}}$$

$$|a + jb| = \sqrt{a^r + b^r}, \quad \tan\varphi = \frac{b}{a}$$

$$\tan \phi = - \frac{2\zeta \frac{\omega}{\omega_n}}{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]}$$

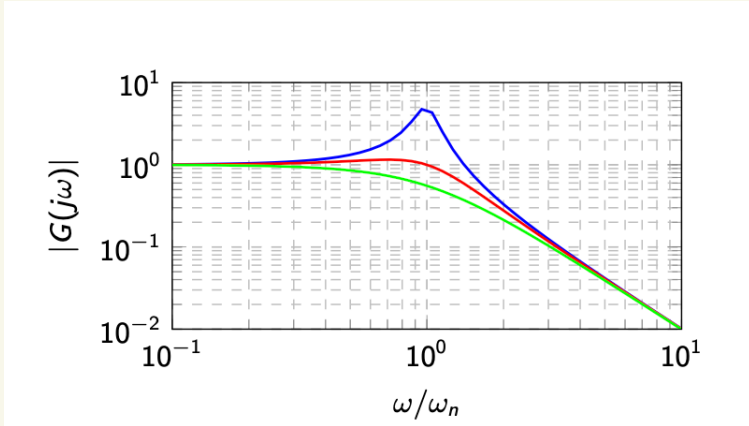


Figure: Amplitude ratio of $G(j\omega)$

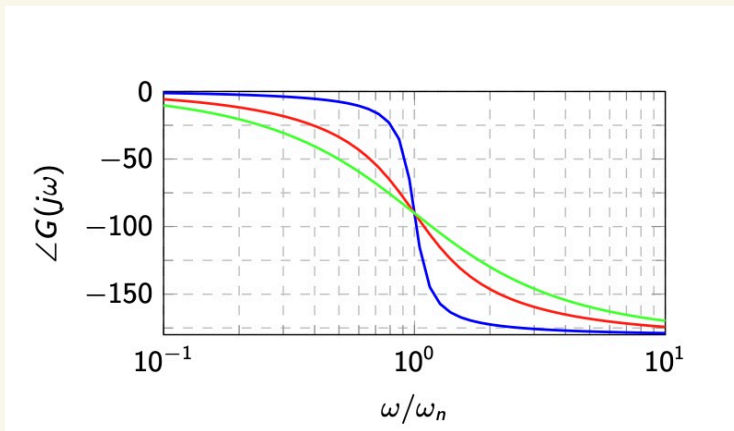


Figure: Phase of $G(j\omega)$ with frequency change.

Approximations:

$$\text{ff } \omega \leq \omega_n \quad |G(j\omega)| = 1$$

$$\text{ff } \omega \geq \omega_n \quad |G(j\omega)| = \left(\frac{\omega}{\omega_n}\right)^{-2}$$

$$\Rightarrow \log |G(j\omega)| = - 2 \log\left(\frac{\omega}{\omega_n}\right)$$

$$\text{ff } \omega = \omega_n, \quad |G(j\omega)| = \frac{1}{2\zeta}$$

Note that at higher frequencies the amplitude is attenuated.

The phase shifts from 0 to -180° . The phase lag increases.

